Here is the complete project documentation for your employee salary calculation program, formatted just like the previous ones!

1. Problem Statement, Objectives, and Assumptions

Problem Statement

Manual payroll processing—incorporating basic salaries, variable components like overtime and incentives, house rent allowances (HRA), standard fixed bonuses, and deductions—is complex, repetitive, and susceptible to mathematical errors. An automated payroll calculation script is required to instantly compute accurate gross and net earnings for employees.

Objectives

- **Automate Payroll Computation:** To seamlessly compute total earnings by combining base pay, allowances, and variable performance incentives.
- **Standardize Allowances & Deductions:** To automatically calculate House Rent Allowance (HRA) as a percentage of the basic pay, apply standard overhead allowances, and compute mandatory deductions.
- **Ensure Financial Transparency:** To provide employees or HR departments with a clear itemized breakdown of HRA, gross salary, deductions, and final take-home (net) salary.

Assumptions

- Financial inputs (basic salary, overtime, incentives) are provided as numeric values.
- House Rent Allowance (HRA) is fixed at 20% of the basic salary.
- A standard fixed allowance of Rs. $1{,}000$ is added to every employee's gross calculation.
- Total deductions are computed at a flat rate of 10% of the gross salary.

2. Algorithm, Flowchart, and Pseudocode

Algorithm

1. **Start** the program.
2. **Input** basic salary, overtime amount, and incentive amount as floating-point numbers.
3. **Calculate HRA:** hra = basic * 0.20.
4. **Calculate Gross Salary:** gross = basic + hra + 1000 + overtime + incentive.
5. **Calculate Deductions:** deduction = gross * 0.10.

6. **Calculate Net Salary:** net = gross - deduction.

7. **Display Results:** Print HRA, Gross Salary, Deduction, and Net Salary amounts.

8. **End** the program.

Flowchart Representation

Plaintext

[START]

    |

    ▼

[Input: basic, overtime, incentive]

    |

    ▼

[hra = basic * 0.20]

    |

    ▼

[gross = basic + hra + 1000 + overtime + incentive]

    |

▼

[deduction = gross * 0.10]

|

▼

[net = gross - deduction]

|

▼

[Print: hra, gross, deduction, net]

|

▼

[END]

Pseudocode

Plaintext

```
BEGIN
    PRINT "Enter basic salary: "
    READ basic
    PRINT "Enter overtime amount: "
    READ overtime
    PRINT "Enter incentive: "
```

```
    READ incentive

    hra = basic * 0.20
    gross = basic + hra + 1000 + overtime + incentive
    deduction = gross * 0.10
    net = gross - deduction

    PRINT "HRA:", hra
    PRINT "Gross Salary:", gross
    PRINT "Deduction:", deduction
    PRINT "Net Salary:", net
END
```

3. Python Source Code, Test Cases, and Outputs

Python Source Code

Python

```
# Employee Salary Calculation System

basic = float(input("Enter basic salary: "))
overtime = float(input("Enter overtime amount: "))
```

```
incentive = float(input("Enter incentive: "))


hra = basic * 0.20
gross = basic + hra + 1000 + overtime + incentive
deduction = gross * 0.10
net = gross - deduction


print("HRA:", hra)
print("Gross Salary:", gross)
print("Deduction:", deduction)
print("Net Salary:", net)
```

Test Cases Table

Test Case ID	Basic (Rs.)	Overtime (Rs.)	Incentive (Rs.)	HRA (Rs.)	Gross (Rs.)	Deduction (Rs.)	Net Salary (Rs.)	Status
TC_01	20000.0	2000.0	3000.0	4000.0	30000.0	3000.0	27000.0	Pass
TC_02	15000.0	1000.0	500.0	3000.0	20500.0	2050.0	18450.0	Pass

TC_03	30000.0	5000.0	4000.0	6000.0	46000.0	4600.0	41400.0	Pass
TC_04	10000.0	0.0	0.0	2000.0	13000.0	1300.0	11700.0	Pass

Simulated Terminal Output

Bash

Enter basic salary: 20000

Enter overtime amount: 2000

Enter incentive: 3000

HRA: 4000.0

Gross Salary: 30000.0

Deduction: 3000.0

Net Salary: 27000.0

4. Conclusion

This Python script effectively streamlines the payroll calculation process by integrating percentage-based allowances, fixed bonuses, performance incentives, and proportional deductions into a single execution workflow. It provides an efficient and reliable programmatic foundation for human resource and accounting software systems.